

GROUP C

REPORT PRODUCTION

**8 December 2025
Engineering Statistics**

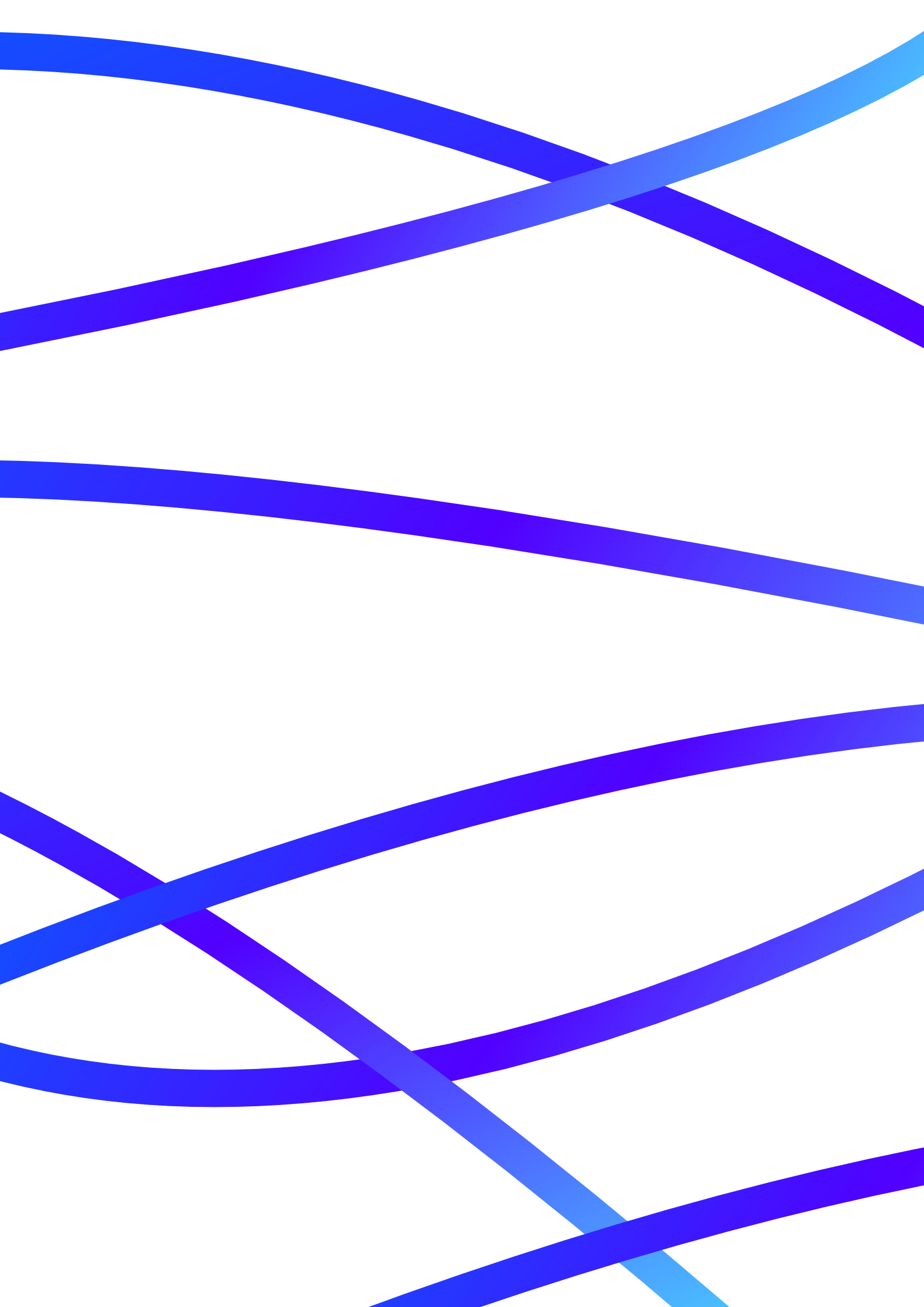


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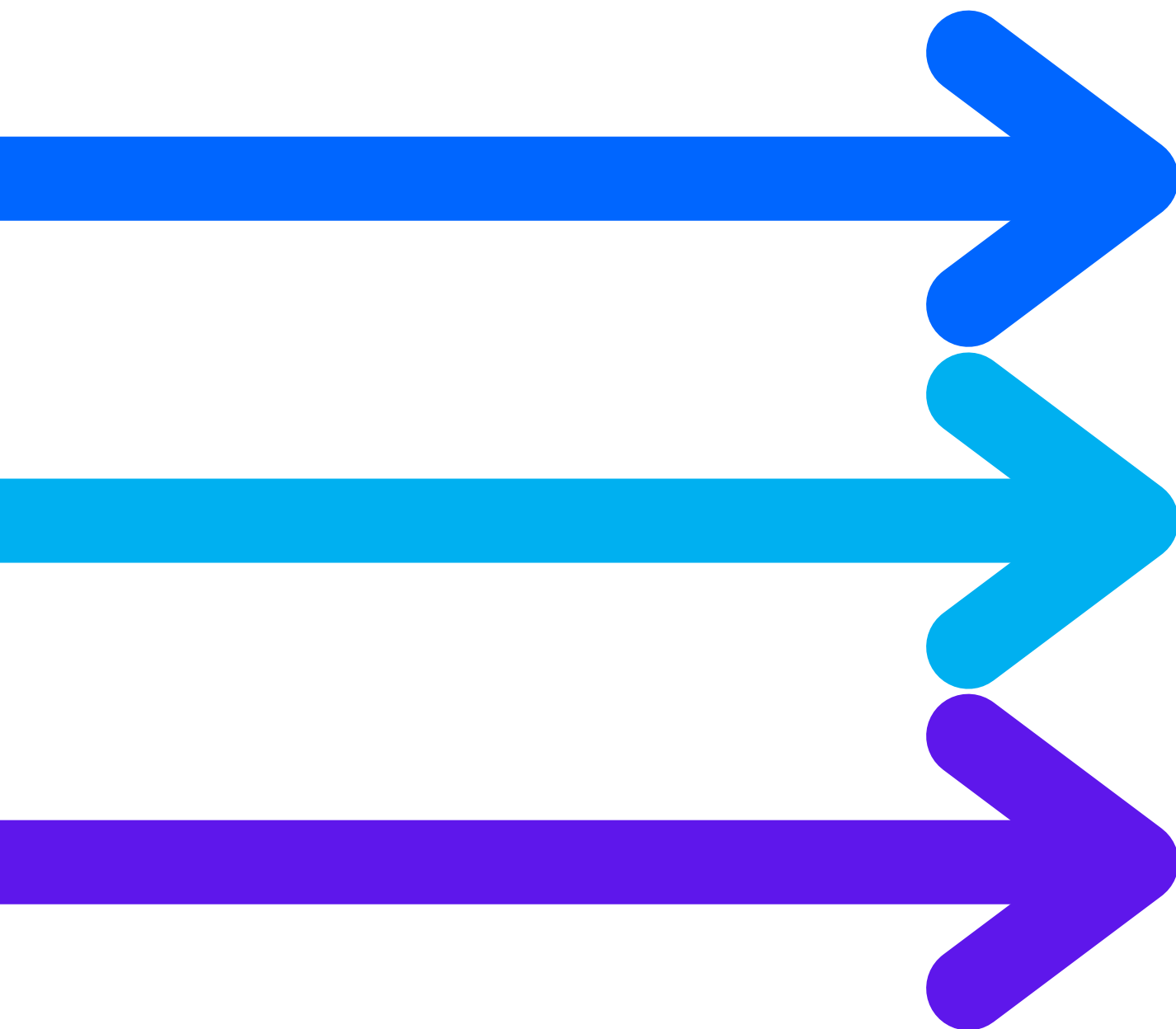




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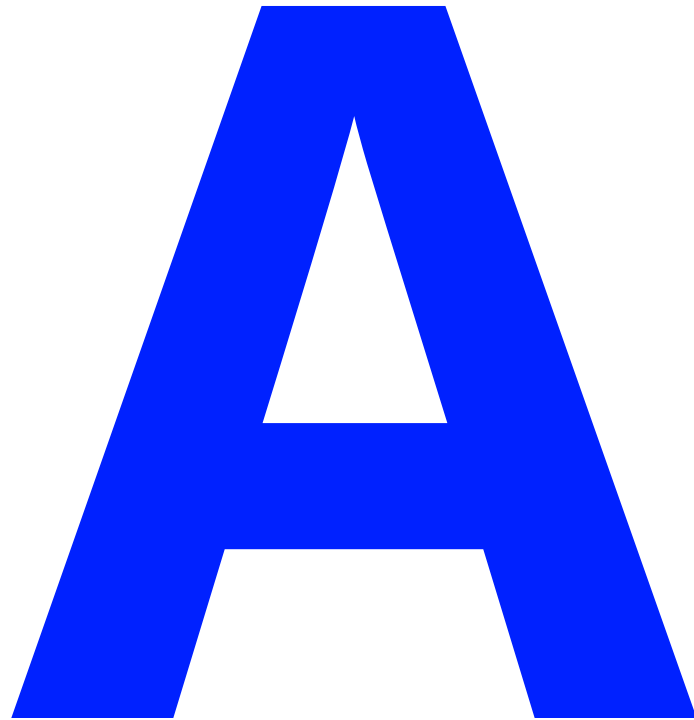
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TEAM MEMBERS DESCRIPTION

Our Names and Student IDs, and Information About the Material



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INFORMATION ABOUT MATERIAL

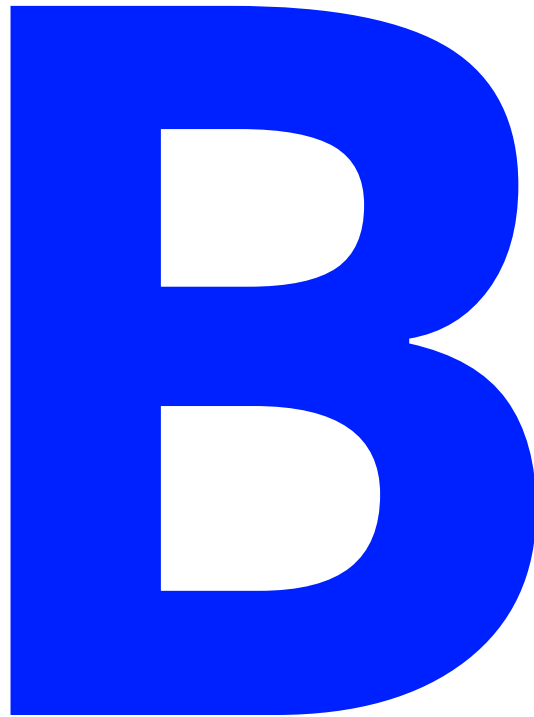
University Name: Taif University

Course Name: Engineering statistics

Instructor's Name: Yazid Al-Zaidi

Year: 2025





CONTENT

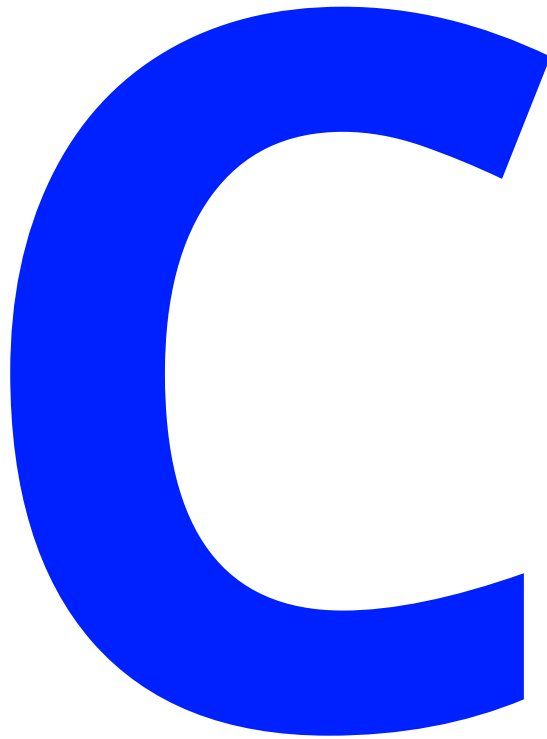
This study analyzes a dataset provided for the Engineering Statistics course, explaining the data, the methods used, and the results with practical insights.

CONTENT

Data is considered one of the most important resources in the modern era. It plays a vital role in understanding various phenomena and supporting decision-making based on scientific evidence. With the increasing volume and diversity of data, analyzing it in an organized manner has become essential to extract useful information and accurate insights that help researchers and decision-makers better understand reality and take appropriate actions.

The purpose of this study is to analyze the dataset provided by Dr. Yazid Al-Zaidi as part of the requirements of the Engineering Statistics course. The team was given this dataset in a ready-to-analyze format, which includes a set of variables and important information that reflects different aspects of the studied phenomenon.

This report highlights several important components, starting with describing the data and clarifying its nature and source, then explaining the statistical methods used in the analysis, and presenting the results through tables and charts. It also includes a discussion of these results and their practical implications. In addition, the report provides clear conclusions based on the statistical evidence obtained, along with recommendations that may benefit future research or related practical applications.



OBJECTIVE

Main goal: To check out the dataset for the Engineering Statistics course and pull out useful info and clear insights that help understand how the production process really works.

OBJECTIVE

1 Understanding and Describing the Data

Looking at the factors affecting 5,000 products in the factory, such as temperature, pressure, speed, and production time. Identifying the type of each variable and understanding it clearly.

2 Applying Statistical Analysis

Using basic statistics to understand averages and how the data is spread.

3 Evaluating Production Quality

Using basic statistics to understand averages and how the data is spread.

4 Studying Relationships and Correlations

Measuring the strength and direction of the relationship between quality and other variables to identify the main influencing factors. Also checking relationships among the operational variables themselves.

5 Time-Based Performance Analysis

Monitoring quality changes over the years to see if it improves, declines, or fluctuates randomly.

6 Conclusions and Recommendations

Identifying the root problems. Providing practical recommendations to improve factory operations or train operators based on statistical results.

01

DATA DESCRIPTION

Includes: Types of Data, Definition and Meaning of Variables, Data Overview

DATA DESCRIPTION

The dataset provided by Dr. Yazid Al-Zaidi for the Engineering Statistics course was used in this study. This data represents 5,000 products in a factory, which are affected by factors such as temperature, speed, pressure, and minimum production time. Quality is the main focus, as it reflects the final product and shows how temperature, pressure, production time, and speed can affect the product either positively or negatively. The dataset was provided to the group in a ready-to-analyze format and will be used for statistical analysis and to draw meaningful conclusions.

Types of Data

1. Categorical Data

- The Product ID (Product_ID)

It represents only the product number and is considered categorical data.

- The Production Line (Production_Line)

It represents the production line, numbered from 1 to 5, and is considered categorical data because they are just labels.

- Team Group (Shift)

It represents the group of employees responsible for the product. There are three employee groups: A, B, and C.

DATA DESCRIPTION

- **The Operation Number (Operator_ID)**

It is the operation number within the manufacturing process, always starting with "OP" followed by the operation number.

- **Machine Identifier (Machine_ID)**

This is the machine number, which differs from one unit to another. Here we have 10 machines, from MACH_01 to MACH_10 in order.

- **Defect Status (Defect_Flag)**

It indicates whether the product failed (BAD) or passed (GOOD).

1. Quantitative Data

- **Date**

This is the production date of each product, ranging from 2010 to 2023, which is the period covered by the data. Here, we will consider the date as a quantitative variable and use it in our report.

- **Temperature (Temperature_C)**

Temperature is one of the factors affecting quality. It represents the heat the product is exposed to during its production process and is given in degrees Celsius.

- **Pressure (Pressure_psi)**

Pressure is one of the most important factors affecting product quality. It is considered a quantitative variable that influences the outcome.

DATA DESCRIPTION

- Speed Revolutions Per Minute (Speed_rpm)

Speed represents the rotation speed that the product is subjected to.

- Production Time Minimum (Production_Time_min)

It is the minimum production time, which gives us the shortest time required for each product.

- Quality (Quality_Score)

Quality is considered the main focus of this report and is the primary variable, measured after the product is produced, not before.

Data Overview:

Variable	Mean	Standard deviation	Minimum value	Maximum Value
Quality	83.37	3.98	68.84	97.84
Temperature	189.04	9.01	157.30	221.30
Pressure	201.46	12.28	159.10	243.90
Speed	1214.40	50.21	1058.60	1368.50
Production Time Minimum	123.70	5.47	106.90	141.60

02

ANALYSIS METHODOLOGY

Includes: Statistical Methods, Software Used for Analysis

ANALYSIS METHODOLOGY

Here is an explanation of the methods we used to analyze the data. First, we made sure the dataset was complete and had no missing values. Then, we performed simple calculations like the mean to understand the general shape of the data. We also used charts and tables to make the results easier and clearer to interpret.

Statistical Methods:

Mean (Average):

We used the mean because it is the foundation for calculating standard deviation, variance, and other operations. It is useful for knowing the typical value in the dataset and understanding the general trend (whether most values are high, medium, or low).

Median:

We used the median because it shows the middle value after arranging the data. It helps, together with the mean, to see if the distribution leans left, right, or is symmetric, giving a clear view of the dataset.

Standard Deviation:

This measures how much the data is spread out. We used standard deviation instead of variance because it gives a more accurate measure of data dispersion, whereas variance is the square of the spread.

ANALYSIS METHODOLOGY

Variance:

Variance measures how spread out the data is, but since it uses squared values, it is less intuitive compared to standard deviation.

Minimum Value:

This is the smallest value in the dataset. It is important for understanding the data range and is useful in decision-making and analysis.

Maximum Value:

This is the largest value in the dataset. It is also important for understanding the data range and helps indicate the highest value reached in the dataset.

Range:

The range is calculated by subtracting the smallest value from the largest value. It helps to understand the spread of the data and allows for comparisons.

Mode:

We calculated the mode and the percentage of "good" and "bad" in the defect flag to check whether we are on the right track or not.

Interquartile Range (IQR):

The IQR measures the true spread of the data without being affected by outliers, because it focuses only on the middle portion of the dataset. It gives a more accurate picture of the range where most values actually lie.

ANALYSIS METHODOLOGY

Correlation Coefficient:

The correlation coefficient measures the strength and direction of the relationship between variables. A value close to 1 indicates a strong positive relationship, while a value close to -1 indicates a strong negative relationship. A value near 0 means there is little or no relationship.

Software Used for Analysis:

We only used Excel for our analysis. The main reason is that it's easy to use and all the data is already there. Plus, it can do many tasks, like calculating the percentage of "good" and "bad" and making a pie chart, all without using any other program.

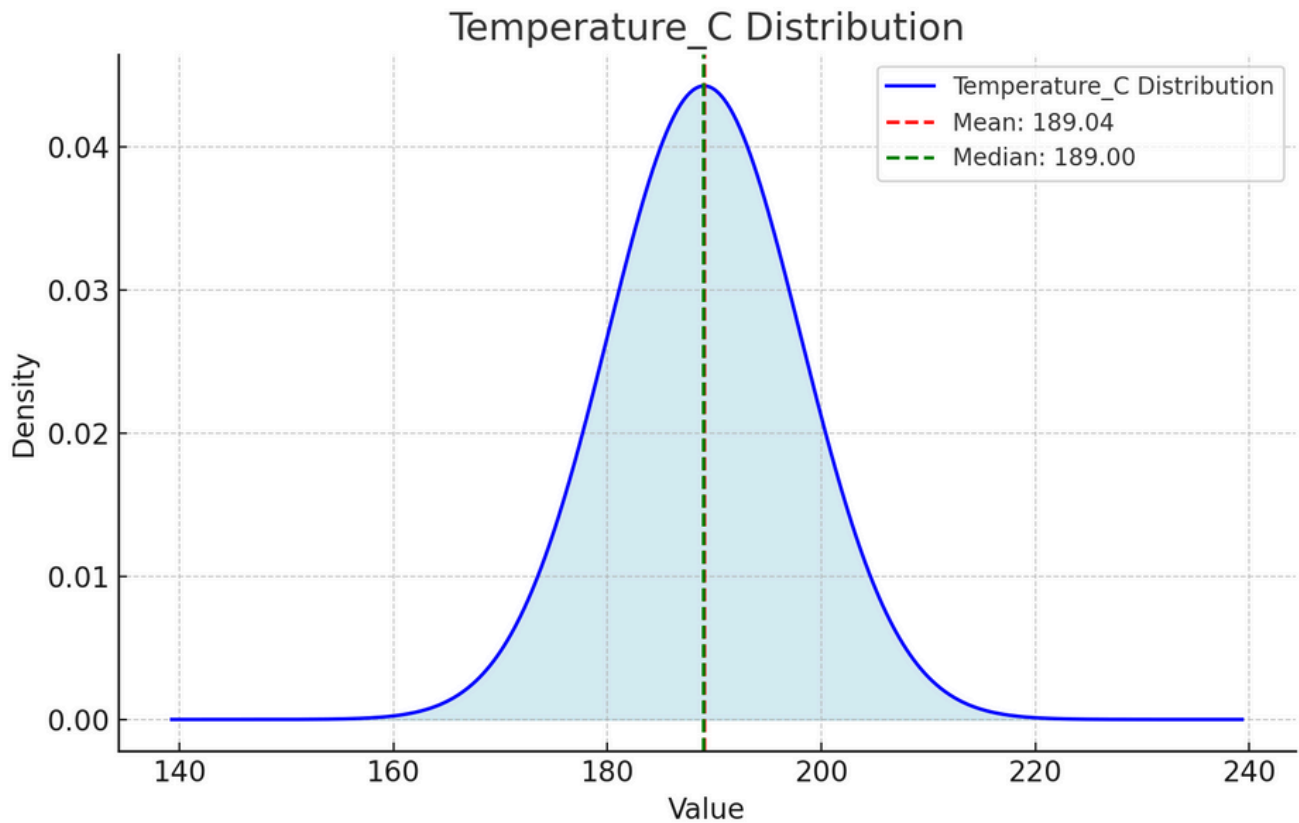
03

STATISTICAL ANALYSIS

Includes: Plotting Temperature, Pressure, Speed, PTM, Quality on a Density Curve, Descriptive Statistics by Defect Flag, Quality Score - Descriptive Statistics by Shift, Correlation Coefficients, Scatter Plot, And Time Series plot.

STATISTICAL ANALYSIS

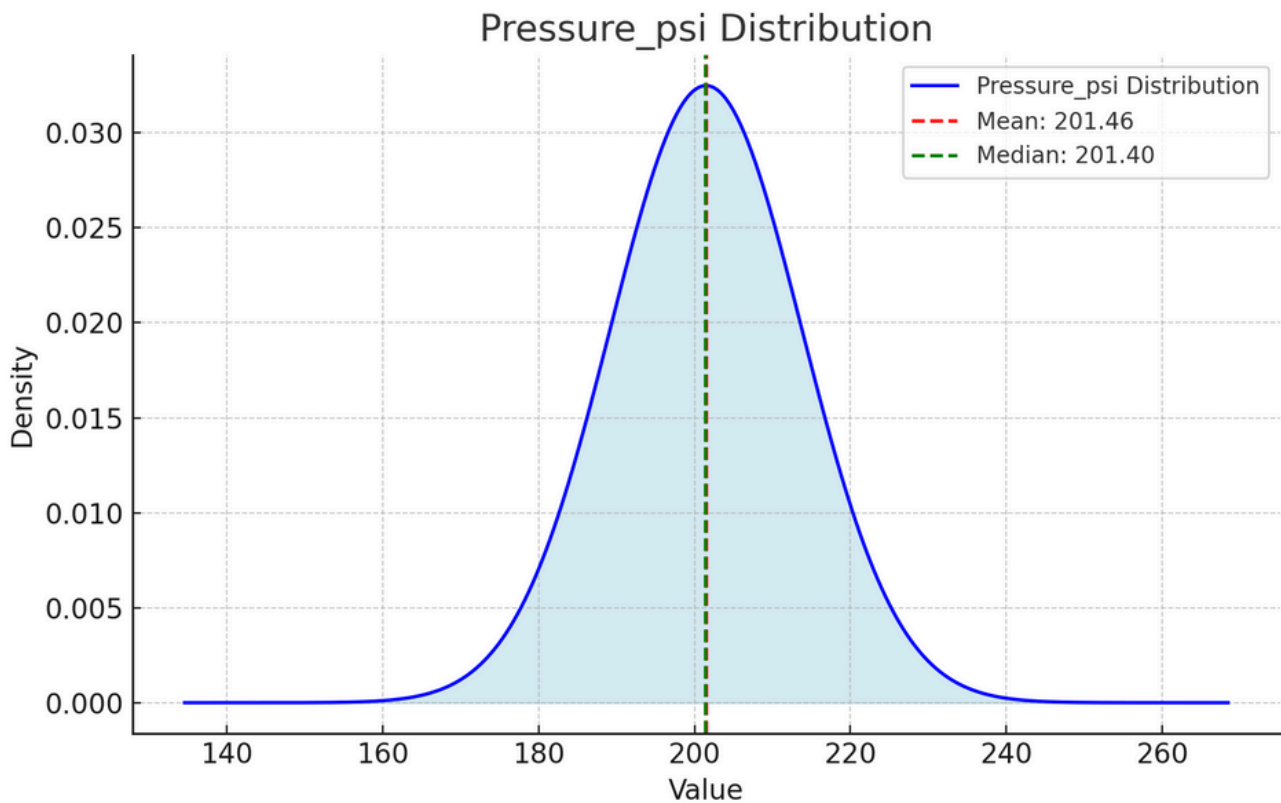
Plotting Temperature on a Density Curve:



Mean	Median	Std	Min	Max
189	189	9	157.3	221.3
<i>Measures of Central Tendency</i>	<i>Measures of Central Tendency</i>	<i>Measures of Dispersion</i>	<i>Measures of Dispersion</i>	<i>Measures of Dispersion</i>

STATISTICAL ANALYSIS

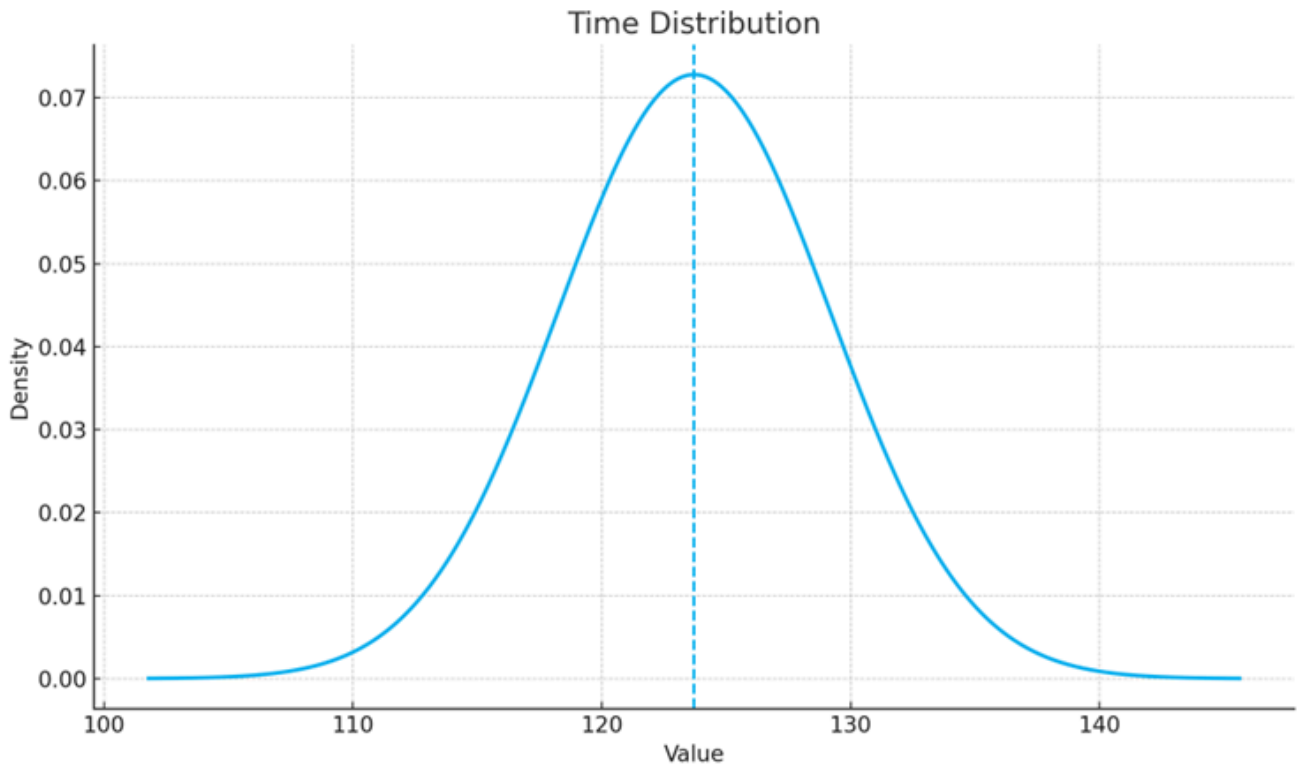
Plotting Pressure on a Density Curve:



Mean	Median	Std	Min	Max
201.4	201.4	12.28	159.1	243.9
Measures of Central Tendency	Measures of Central Tendency	Measures of Dispersion	Measures of Dispersion	Measures of Dispersion

STATISTICAL ANALYSIS

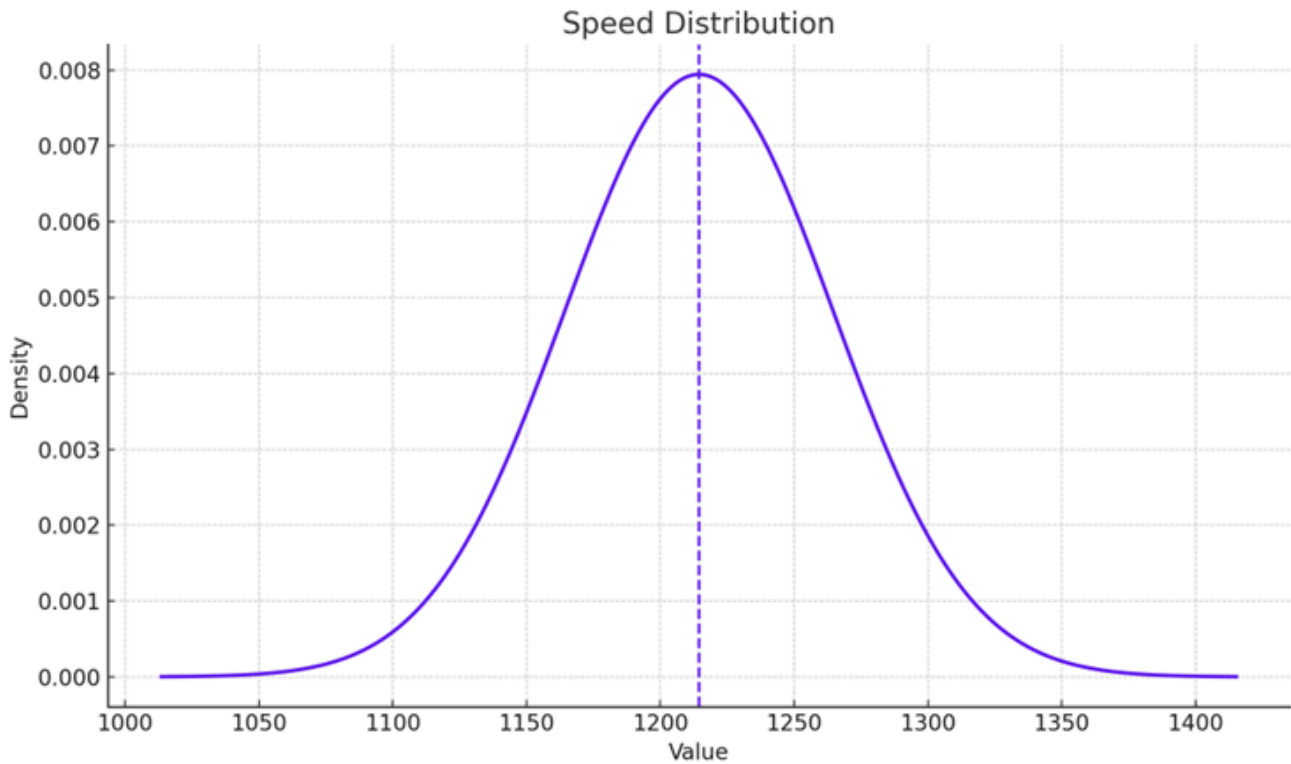
Plotting Production Time Minimum on a Density Curve :



Mean	Median	Std	Min	Max
123.7	123.7	5.47	106.9	141.6
<i>Measures of Central Tendency</i>	<i>Measures of Central Tendency</i>	<i>Measures of Dispersion</i>	<i>Measures of Dispersion</i>	<i>Measures of Dispersion</i>

STATISTICAL ANALYSIS

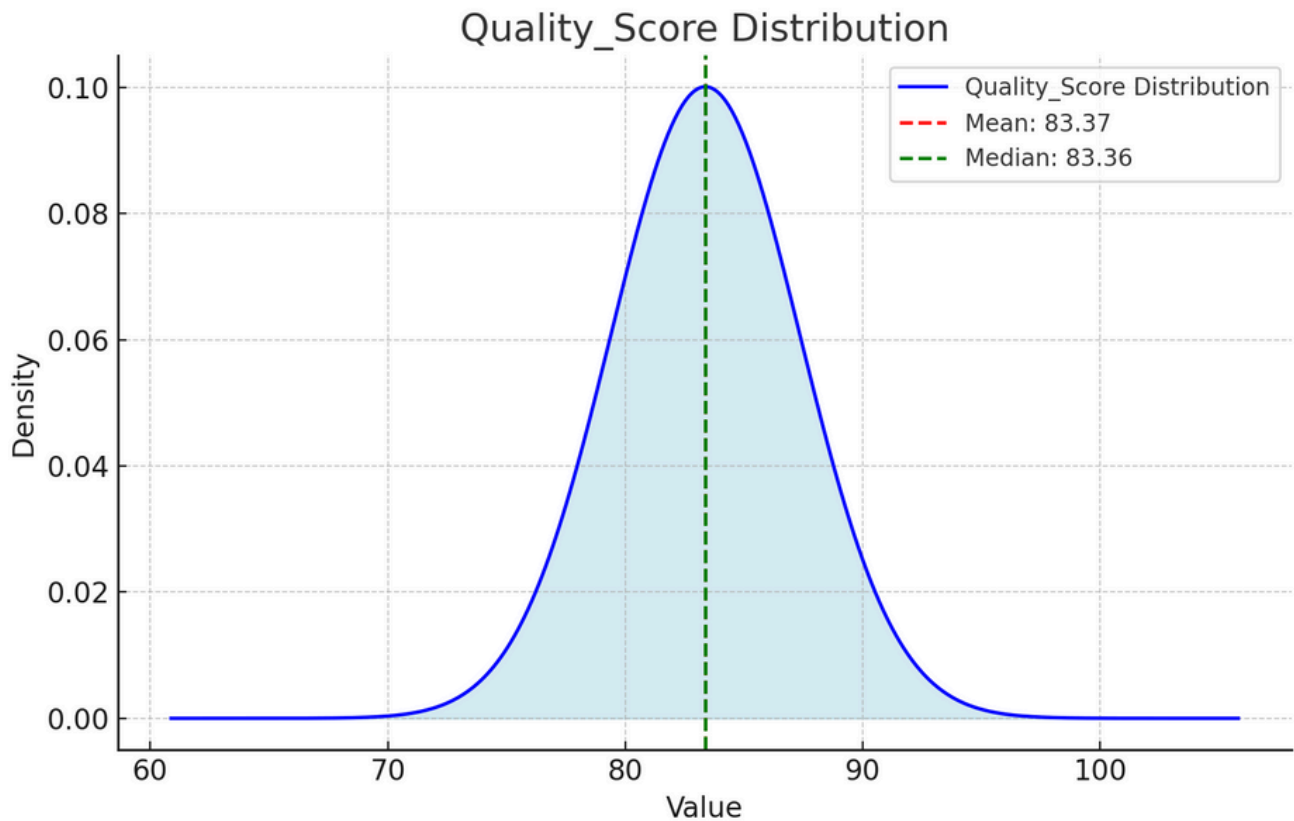
Plotting Speed on a Density Curve:



Mean	Median	Std	Min	Max
1214	1214	50.21	1058	1368
<i>Measures of Central Tendency</i>	<i>Measures of Central Tendency</i>	<i>Measures of Dispersion</i>	<i>Measures of Dispersion</i>	<i>Measures of Dispersion</i>

STATISTICAL ANALYSIS

Plotting Quality on a Density Curve:



Mean	Median	Std	Min	Max
83.3	83.3	3.9	68.8	97.8
Measures of Central Tendency	Measures of Central Tendency	Measures of Dispersion	Measures of Dispersion	Measures of Dispersion

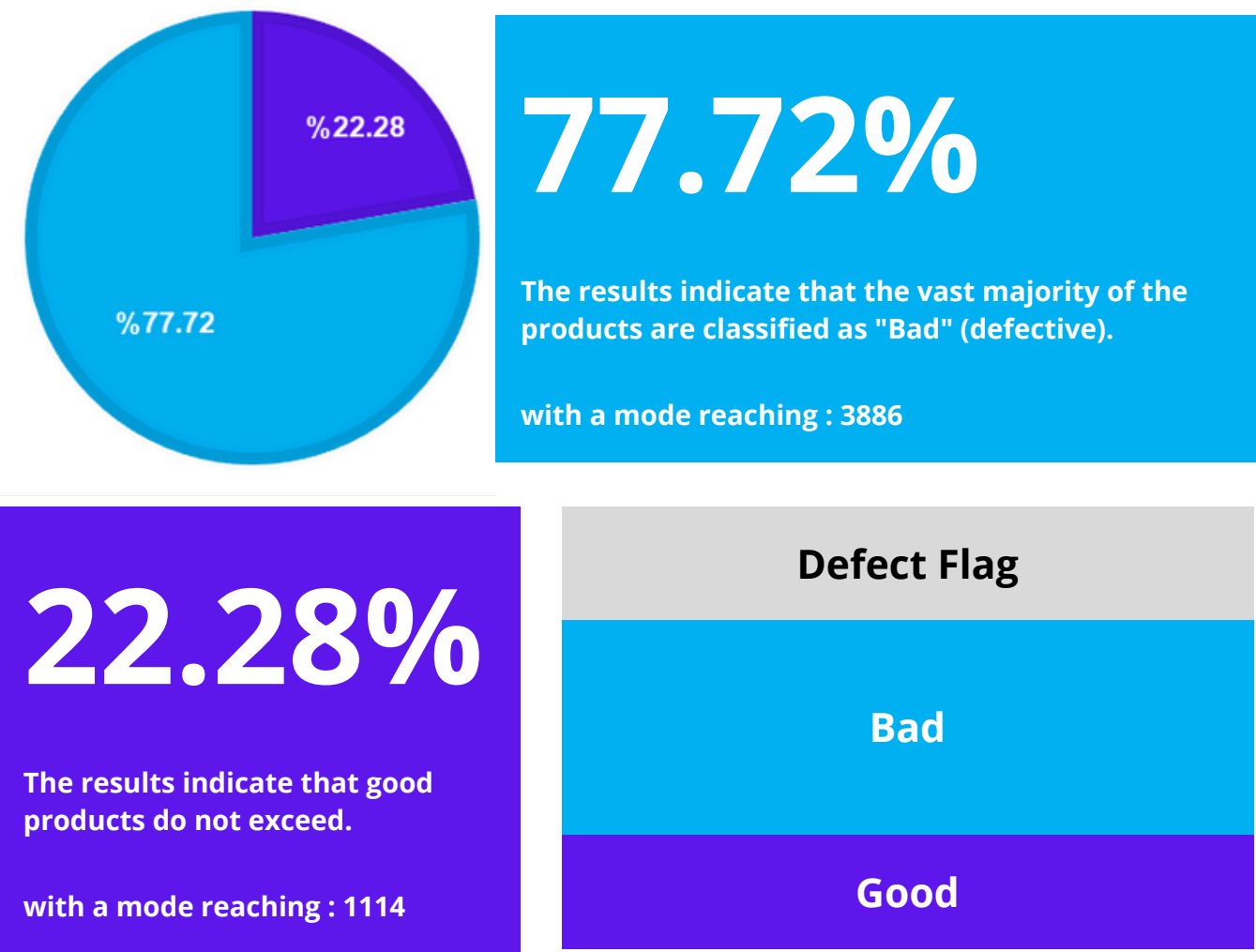
STATISTICAL ANALYSIS

The average quality is approximately 83.37 with a standard deviation of 3.98, indicating a distribution centered around this mean. This reflects stability in production quality. Temperature and pressure appear to be within stable ranges, with means of approximately 189 and 201 respectively. It is recommended to continue monitoring to ensure these variables remain within optimal limits.

From looking at the data and checking the statistical values for speed and time, it shows that the average speed is about 1214.407, while the average time is 123.708. This means there's a clear difference between the two variables. Also, the mean and median are very close for both speed and time, which tells us the data is pretty balanced and doesn't have big outliers or weird values. The standard deviation also shows that the variation is small, meaning the values are stable and the results are reliable. For the range, the difference between the highest and lowest values shows some variation, but it's still within normal and acceptable limits. There's nothing extreme that would affect the overall understanding of the data. So overall, the data is consistent and shows stable performance for both speed and time, with room for improvement to reach even better efficiency and quality in the future.

STATISTICAL ANALYSIS

Descriptive Statistics by Defect Flag:

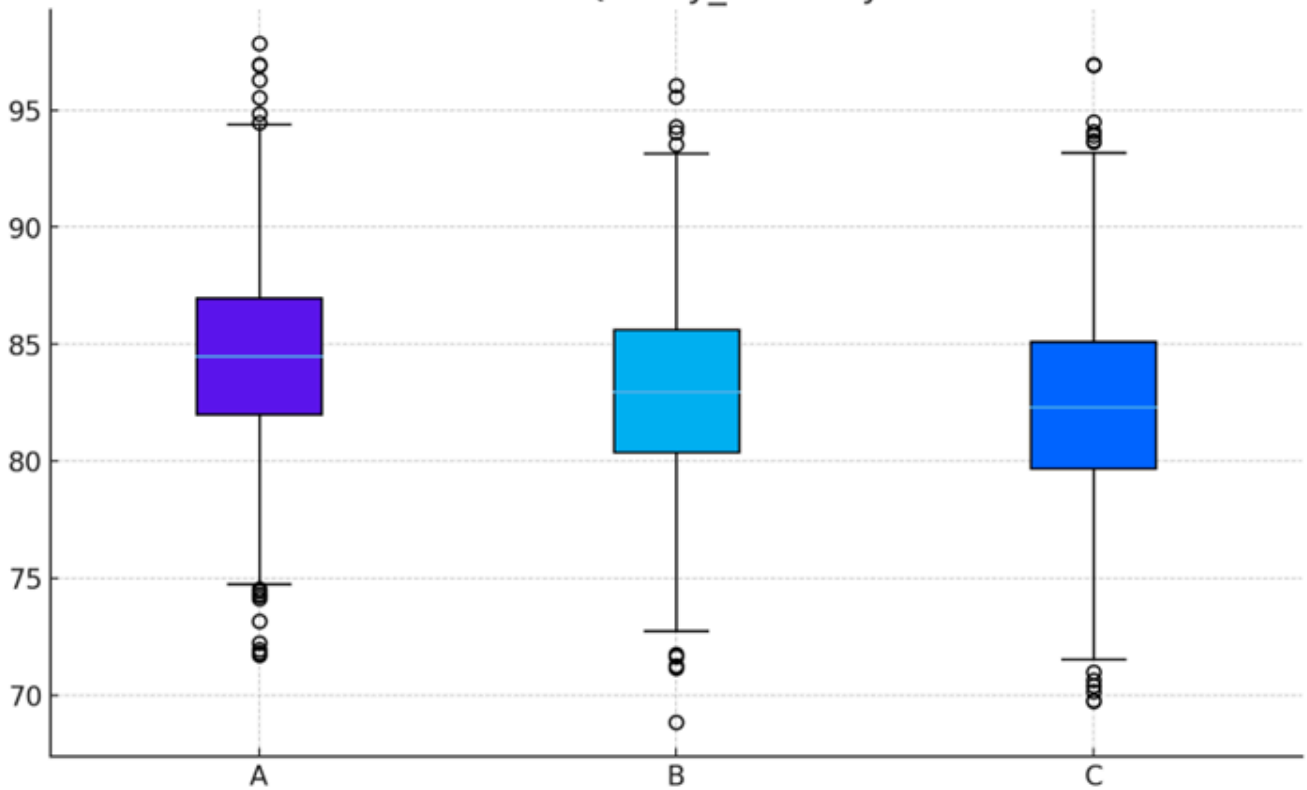


Quality Score - Descriptive Statistics by Shift:

Mean	Shift A	Mean	Shift B	Mean	Shift C
84.5		83		82.41	
Median	Shift A	Median	Shift B	Median	Shift C
84.5		82.9		82.3	
Var	Shift A	Var	Shift B	Var	Shift C
14.8		14.7		15.95	

STATISTICAL ANALYSIS

Box Plot of Quality_Score by Shift



Std	Shift A	Std	Shift B	Std	Shift C
3.8		3.8		4	
Range	Shift A	Range	Shift B	Range	Shift C
29		28.96		28.96	
IQR	Shift A	IQR	Shift B	IQR	Shift C
5.28		5.28		5.28	
Percentage of Bad	Shift A	Percentage of Bad	Shift B	Percentage of Bad	Shift C
79%		78%		76%	
Number of Bad	Shift A	Number of Bad	Shift B	Number of Bad	Shift C
1403		1405		1078	

STATISTICAL ANALYSIS

1. Stability & Consistency (Rock Solid)

The headline here? The process is locked in. It's crazy how consistent things are across every single shift.

- **IQR (The Spread):** It's a dead heat. Literally identical at 5.28 across the board. The middle 50% of your data is behaving exactly the same way, no matter the shift.
- **Range & Std Dev:** Basically twins. The Range is neck-and-neck (~29 vs 28.96), and the Standard Deviation is super tight (3.8 vs 4).
- **The Takeaway:** There's no "shift drama" here. No one is reinventing the wheel. The machines and the SOPs are running on cookie-cutter consistency.

2. Quality Scores (The Box Plot Vibes)

- **Shift A:** Looks like they're leading the pack. Their median score (that blue line) is edging out the others.
- **Shift C:** They're trailing a bit; their median is the lowest of the bunch.

Outliers: Every shift has some weird spikes—those little dots above and below the box. That means you've got some random cases of super high quality and some total duds happening on every shift.

STATISTICAL ANALYSIS

3. Defect Rates (The Plot Twist)

- **The "Bad" Stats:** Shift C is actually the MVP. They've got the lowest "Bad" percentage (76%) and way fewer bad units (1,078) compared to A and B (who are pushing 1,400). Shift A is actually the worst here with a 79% bad rate.
- **The Elephant in the Room:** Real talk, though? If the "Bad" rate is over 75% everywhere, we've got a massive structural red flag. The whole process might be broken, not just a specific shift—unless "Bad" just means some minor classification thing. But if it means "Scrap," we're in trouble.

Correlation Coefficients:

DATA	Quality	Temperature	Pressure	PTM	Speed
Quality	1.0	-0.213	-0.13	0.007	-0.013
Temperature	-0.213	1.0	0.584	-0.016	0.015
Pressure	-0.13	0.584	1.0	-0.021	0.02
PTM	0.007	-0.016	-0.021	1.0	-0.933
Speed	-0.013	0.015	0.020	-0.933	1.0

STATISTICAL ANALYSIS

This table shows the correlation coefficients between Quality, Temperature, Pressure, PTM, and Speed. Looking at the numbers, we can say:

1. Quality vs the other variables:

- Correlation between Quality & Temperature = -0.213 weak negative relationship.
- Correlation between Quality & Pressure = -0.13 very weak negative.
- Correlation between Quality & PTM = 0.007 basically no relationship.
- Correlation between Quality & Speed = -0.013 basically no relationship.
- Conclusion: Quality doesn't really depend on any of these.

2. Speed vs PTM:

- Correlation = -0.933 really strong negative relationship.
- Conclusion: The faster the speed, the lower the production time, and vice versa.

3. Temperature vs Pressure:

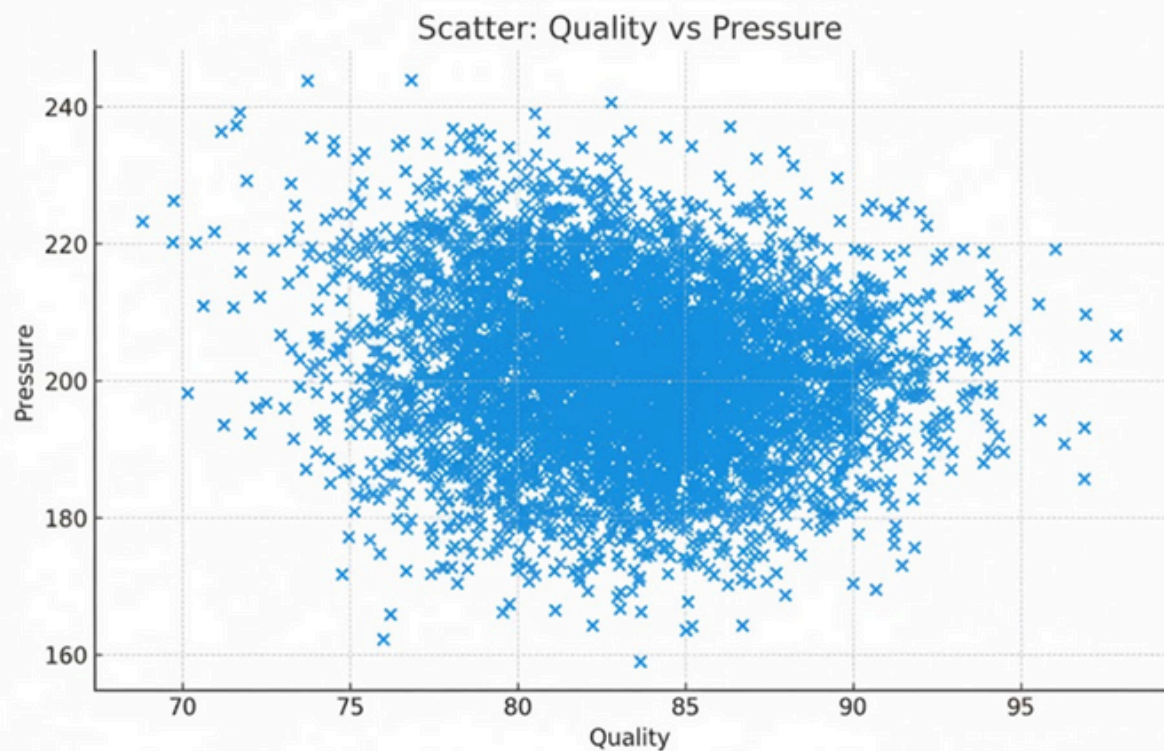
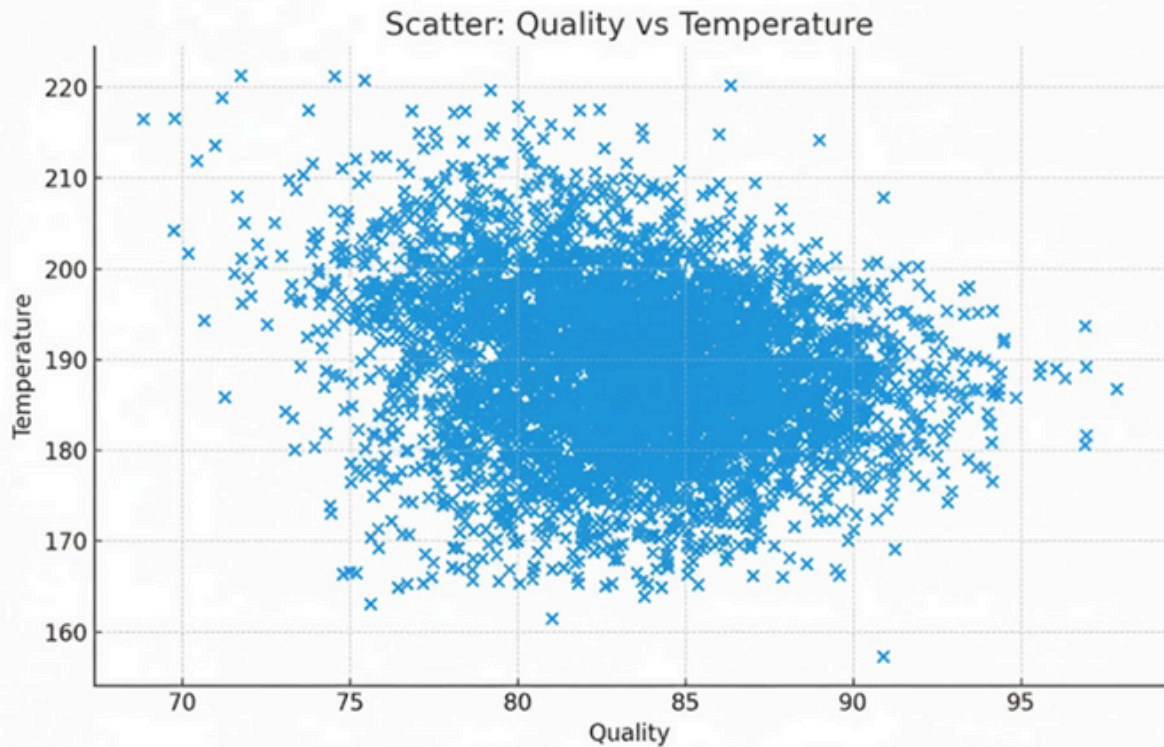
- Correlation = 0.584 moderate positive relationship.
- Conclusion: Higher temperature usually comes with higher pressure.

4. Other relationships:

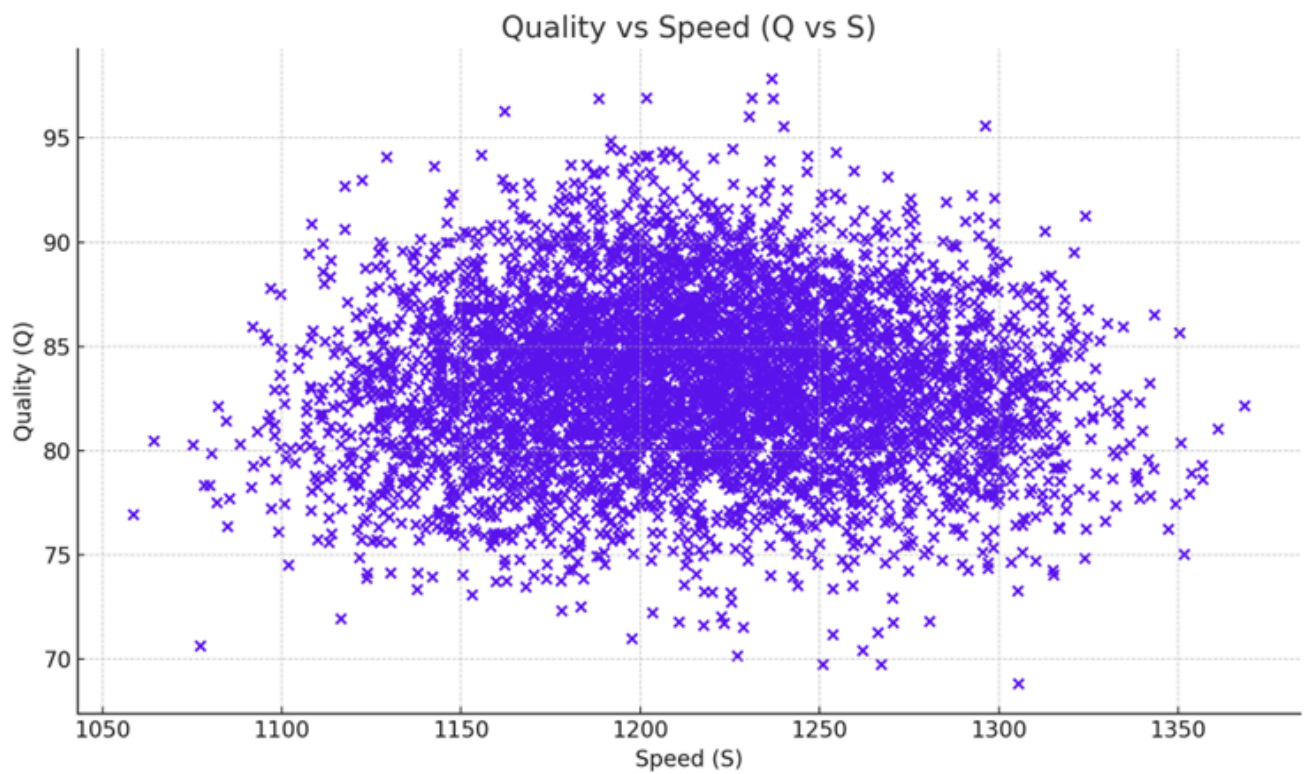
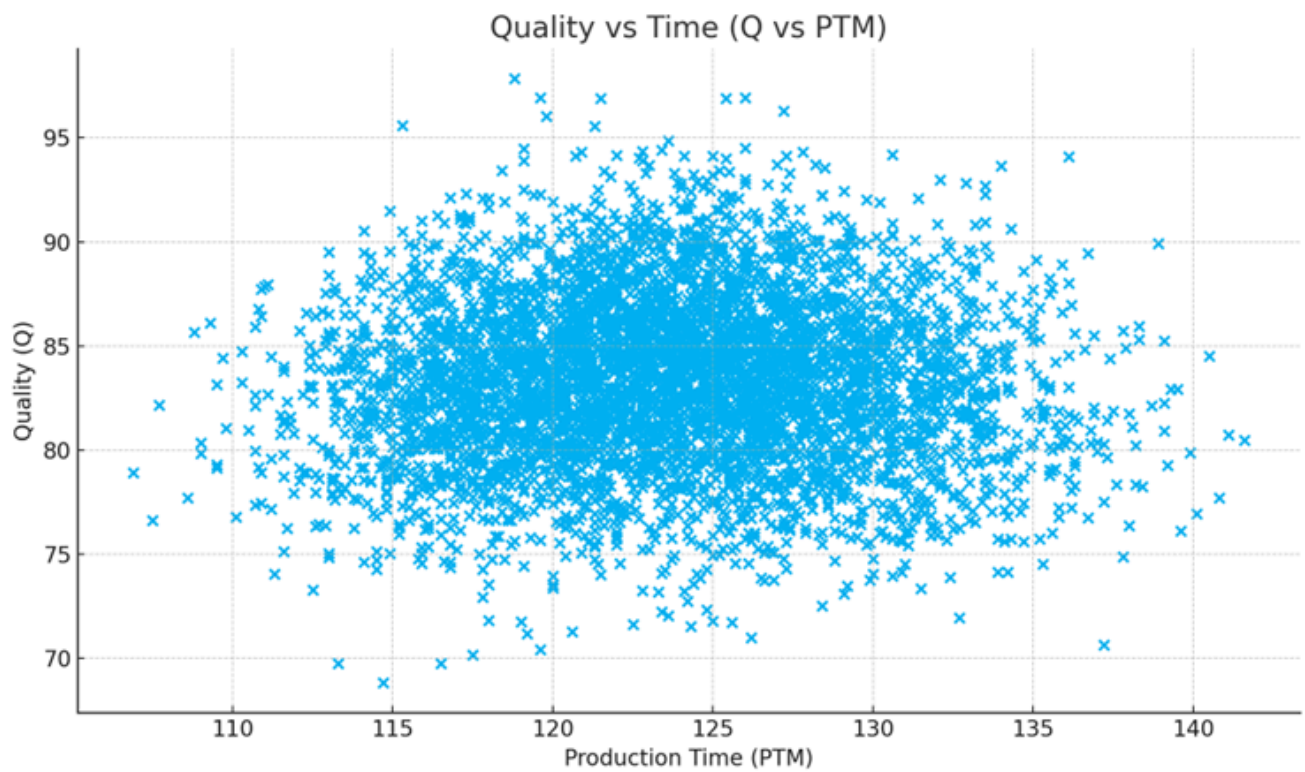
- Most other correlations are super weak, so these variables aren't really linked linearly.

STATISTICAL ANALYSIS

Scatter Plot:



STATISTICAL ANALYSIS



STATISTICAL ANALYSIS

Overall conclusion: There's no clear correlation.

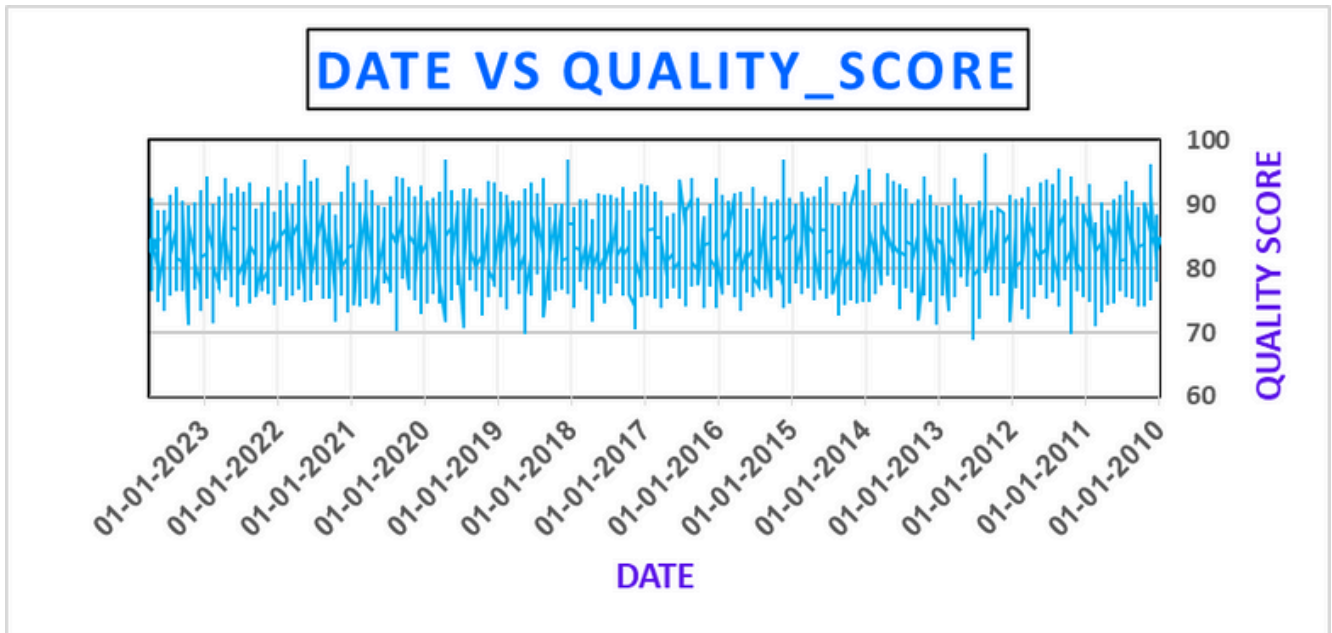
Here's the breakdown for each chart:

- **Quality vs Pressure:** The points are scattered randomly like a circular cloud. No clear upward or downward trend, which means changes in pressure (within this range) don't directly or predictably affect quality.
- **Quality vs Temperature:** Same random pattern here. Temperature alone doesn't seem to be a decisive factor in determining quality in this dataset.
- **Quality vs Production Time Minimum:** Points are all over the place with no clear linear trend. Longer or shorter production time doesn't really relate to better or worse quality.
- **Quality vs Speed:** Even though the points are dense, they're still randomly distributed with no clear pattern linking speed to quality.

The variables are independent the four factors (pressure, temperature, production time minimum, speed) within the current operational ranges don't have a direct role in causing quality fluctuations.

STATISTICAL ANALYSIS

Time Series plot:



Based on the line chart showing Date vs Quality Score, here's the takeaway:

Stable average, but high volatility.

- **No trend:** Over the long period (2010–2023), the chart doesn't show any upward (improvement) or downward (decline) trend. The average quality basically “stays in place” throughout these years.
- **High fluctuation (High Noise):** The line jumps up and down sharply and constantly (roughly between 70 and 95). This means the process output isn't perfectly steady and is affected by daily changing factors causing this “noise” in the results.
- **Time isn't a factor:** Time passing hasn't changed the quality level. The process repeats the same performance and the same level of variation year after year.

04

CONCLUSIONS

Includes: Statistical Methods, Software Used for Analysis

CONCLUSIONS

After going through the dataset and analyzing everything in detail, we ended up with several important points that give us a clear picture of what's happening in the production process. First of all, the quality level is mostly stable overall, and it doesn't really change much over the years. This means the process is running at the same level without any real improvement or decline. However, we did notice a lot of day-to-day fluctuations in the quality scores, which shows that the process is affected by different factors that change from day to day, causing a lot of "noise" in the output. We also found that quality has no strong linear relationship with temperature, pressure, speed, or production time. So within the current operating ranges, none of these variables are directly increasing or decreasing the quality. One of the strongest findings was the very strong negative correlation between speed and production time. The faster the speed, the shorter the production time — which makes perfect sense. But the biggest issue we discovered was the extremely high defect rate. Across all shifts, more than 75% of the products are labeled as "Bad," which means the problem isn't coming from a specific team or shift — it's most likely a process-level issue or something related to how quality is being measured.

CONCLUSIONS

The time-series analysis also showed that quality doesn't get better or worse over the years. It stays at the same level from 2010 to 2023, just with constant fluctuations. In short, the process is running, but it's not really improving.

Recommendations:

1. Reevaluate the entire production process, because the defect rate is extremely high and needs to be understood and fixed.
2. Work on reducing the daily fluctuations in quality, since the high volatility suggests that some factors are not being controlled properly.
3. Monitor temperature and pressure more closely, as they have a moderate relationship and could impact stability if they go out of range.
4. Use the strong relationship between speed and production time to optimize workflow and improve efficiency without hurting quality.
5. Apply continuous monitoring and regular statistical analysis to identify issues early instead of waiting for them to pile up.
6. Review the quality classification criteria, because having such a high percentage of "Bad" products may mean the evaluation method needs adjustment or clarification.

LEVEL 3

2

0

2

5

GROUP C